# Dale Automation

## Training - Oven Maintenance & Operation

### Inspection Details

Report #: Dale Automation 2793

Location:

Inspection Date: 15-Oct-2026 07:51 PM

Inspector Name: Sean Frost

### Asset Details

Asset Category : 8000 LPH Oven

Asset Name : Blue Ribbon Middelburg

Asset # : DSS 403-5

Meter Reading: 50 Hours

Model/Year: Dale Bakermation

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**1. Main Oven Maintenance**

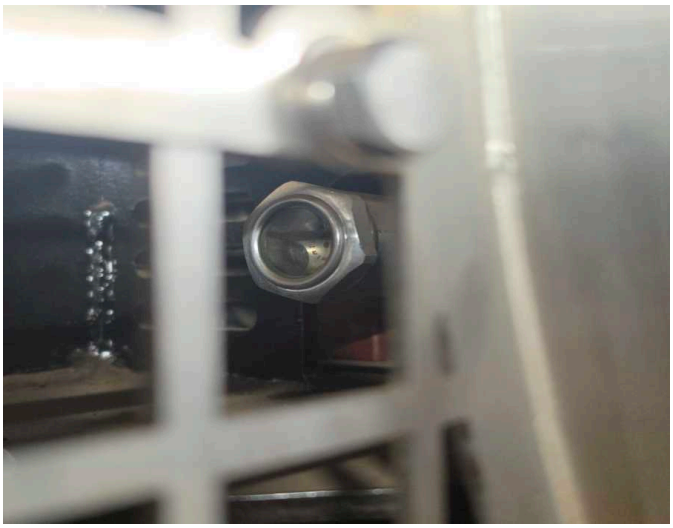
| SN | Inspection Item | Note | Good | Repair | Replace | NA |
|---|---|---|---|---|---|---|
| 1 | Circulation Fans | - V-Belt tensioning method.<br>- Oil Change (3-4 Monthly) Habotec FG46.<br>- Oil quality inspection & Sampling.<br>- Vibration analysis during operation.<br>- Temperature monitoring.<br>- Routine inspection on bearings, stabiliser arms, impellor, fins. | ✓ | ○ | ○ | ○ |
| 2 | Exhaust Fans | - Operation of belimo unit & exhaust damper.<br>- Grease belimo shaft bearings.<br>- Inspect cams & limit switches.<br>- Inspect damper blades.<br>- Vibration & leak test during operation. | ✓ | ○ | ○ | ○ |
| 3 | Chain Grease & Chain Inspection | - Routine greasing of main conveyor chain.<br>- Inspection of rollers are turning freely.<br>- Inspect pins/chain links for signs of wear.<br>- Inspect chain rails for signs of wear.<br>- High torque current limit for grease warning. | ✓ | ○ | ○ | ○ |
| 4 | Main Drive Gearmotor | - Inspect mounting bolts for tightness.<br>- Nosie/vibration inspection during operation.<br>- Oil sampling. Oil change 6-12 monthly.<br>- Drive coupling inspection. | ✓ | ○ | ○ | ○ |
| 5 | Drive & Idle bearings | - Greasing & inspection of main drive bearings, seals, housing tighntess. (XP FG2 grease).<br>- Inspection of idle bearings (Sealed for life).<br>- Inspection for noise & vibration during operation.<br>- Conveyor tension mechanism on idle shaft. | ✓ | ○ | ○ | ○ |
| 6 | Flame Tube Inspection | Inspect flame tubes for signms of degradation, fatigue, corrosion/errosion. | ✓ | ○ | ○ | ○ |

| | | | | | | |
|---|---|---|---|---|---|---|
| 7 | Burner Service & Inspection | Routine service of burners to be conducted by qualified professional. | ✓ | ○ | ○ | ○ |
| 8 | Oven Internal Inspection | Inspection of internal sheeting, pressure chambers, distribution chambers, fan ducting, internal welding. | ✓ | ○ | ○ | ○ |
| 9 | Inspection Door | - Inspect door seal condition, signs of leakage.<br>- Inspect door locks for tightness. | ✓ | ○ | ○ | ○ |
| 10 | Manual Damper Inspection | - Routine actuation of pressure chamber manual dampers. | ✓ | ○ | ○ | ○ |
| 11 | Routine Safety Circuit Test | - E-Stops<br>- Pull chord<br>- Burner E-Stops<br>- Over temperature controllers<br>- Purge circuits | ✓ | ○ | ○ | ○ |
| 12 | Isolation & Lockout Procedures | - Power isolation & lockout procedures to be followed when working on equipment.<br>- "Buddy" system to be adhered to if any works are conducted inside the oven chambers.<br>- General safe working procedures<br>- Loader & Unloader lockout<br>- Safety cages, safety panels, panel safety switches. | ✓ | ○ | ○ | ○ |

## 1. Circulation Fans

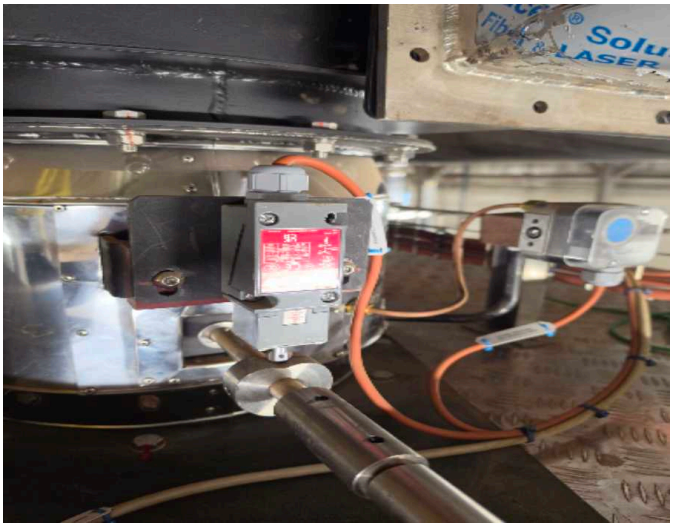






2. Exhaust Fans

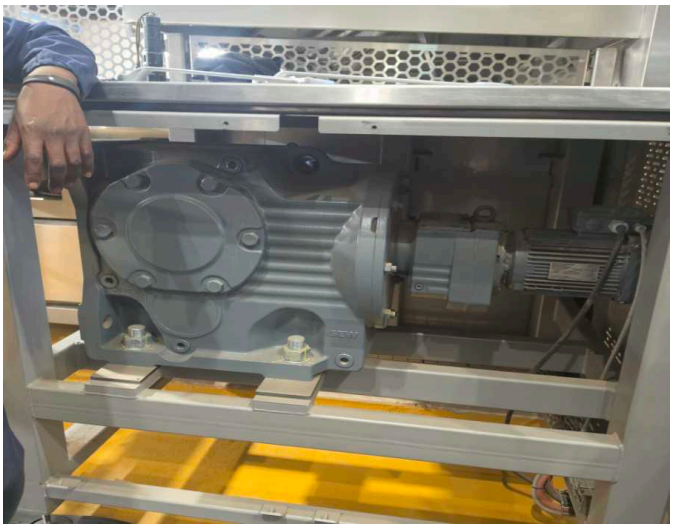


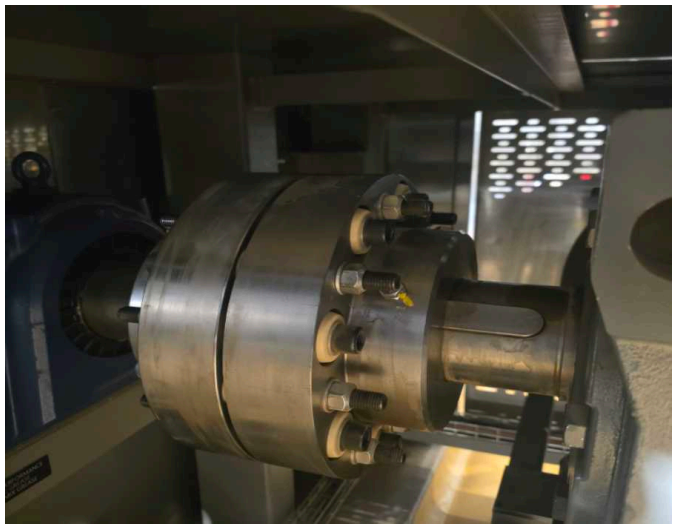


3. Chain Grease & Chain Inspection



4. Main Drive Gearmotor





5. Drive & Idle bearings




6. Flame Tube Inspection



8. Oven Internal Inspection





9. Inspection Door



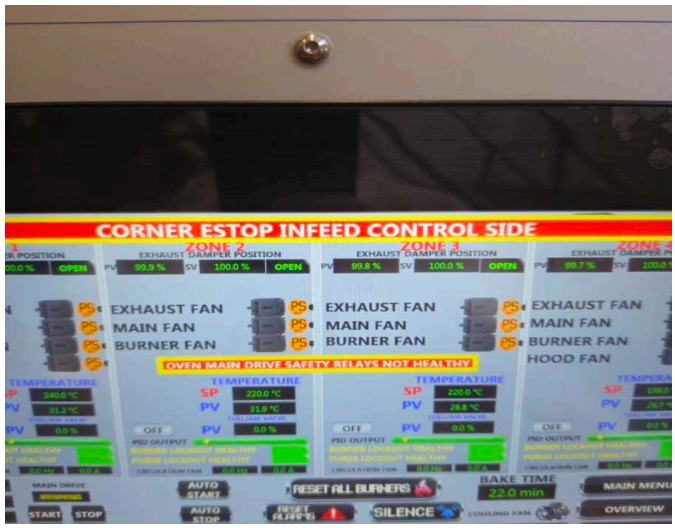


10. Manual Damper Inspection



11. Routine Safety Circuit Test






## 12. Isolation & Lockout Procedures





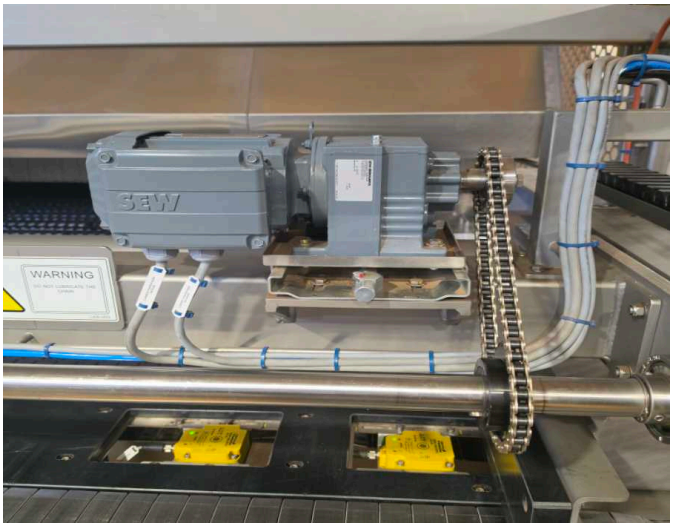


## 2. Oven Loader Maintenance

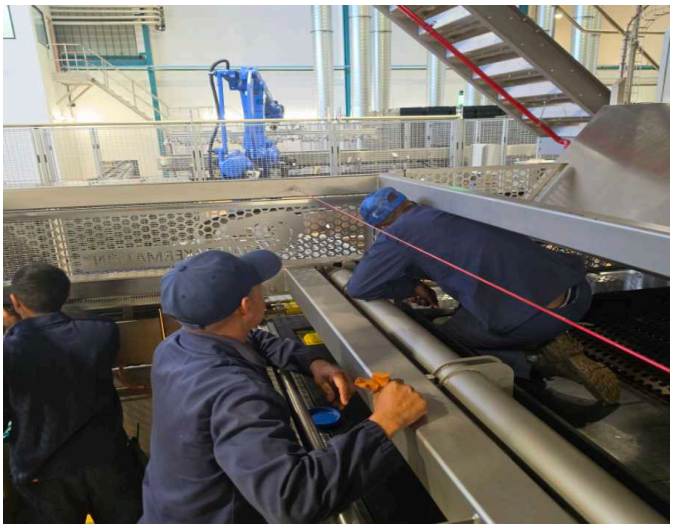
| SN | Inspection Item | Note | Good | Repair | Replace | NA |
|---|---|---|---|---|---|---|

| | | | | | | |
|---|---|---|---|---|---|---|
| 13 | Loader Gearmotors | - 6-12 monthly gearbox oil change (Pusher & Cross Conv).<br>- Inspect mountings & torque arm bushes.<br>- Noise & vibration Inspection. | ✓ | ○ | ○ | ○ |
| 14 | Rack & Pinion | - Routine clean & re-grease<br>- Inspection of rack & spur teeth for wear & smooth engagement.<br>- Inspection of support rollers. | ✓ | ○ | ○ | ○ |
| 15 | Loader Bearings & Drive Coupling | - Pusher arm, driveshaft, cross conveyor bearing re-grease (PF FG2).<br>- Inspection of drive coupling. | ✓ | ○ | ○ | ○ |
| 16 | Loader Pusher Chain & Drive Sprockets | - Chain tension, sprocket tooth inspection for wear, alignment. | ✓ | ○ | ○ | ○ |
| 17 | Loader Pneumatic Arm | - Inspect mechanical end stops.<br>- Inspect position proxies.<br>- Inspect pneumatic cylinders (Stroke, flow control, rod-eye, mounts). | ✓ | ○ | ○ | ○ |
| 18 | Cross Conveyor | - Slat belt & wearstrip inspection.<br>- Drive & idle sprocket inspection.<br>- Roller inspection.<br>- Bearing inspection & re-grease. | ✓ | ○ | ○ | ○ |
| 19 | Pneumatic Inspection | - Inspect for leaks<br>- Clean ASU filter, bowl, condensate trap. | ✓ | ○ | ○ | ○ |
| 20 | Electrical Inspection | - Inspect loader batch proxies, terminal position proxies, pusher position proxies, encoder cables, cable igus. | ✓ | ○ | ○ | ○ |
| 21 | Idle Terminal Anchors | Inspect terminal anchors for tightness. | ✓ | ○ | ○ | ○ |

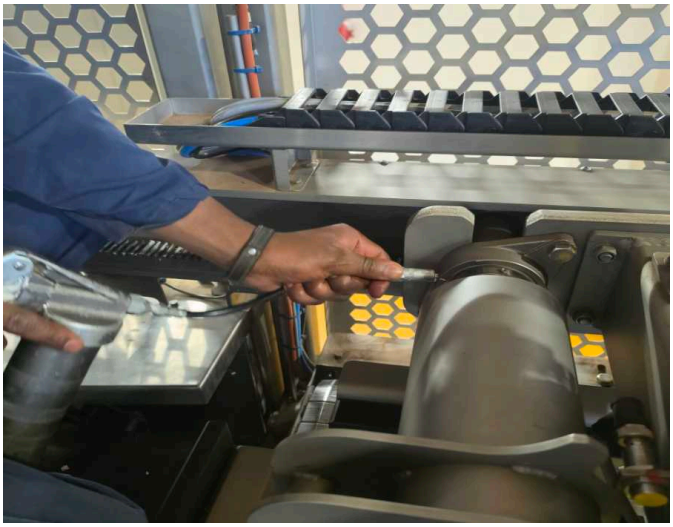
## 13. Loader Gearmotors

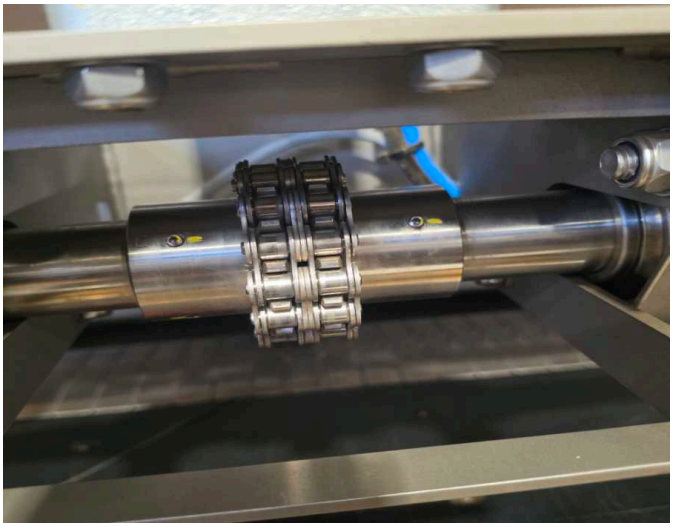



## 14. Rack & Pinion



15. Loader Bearings & Drive Coupling



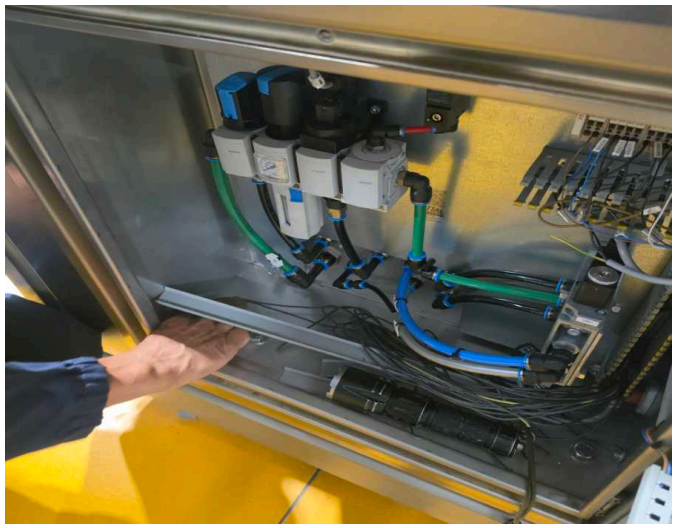


17. Loader Pneumatic Arm

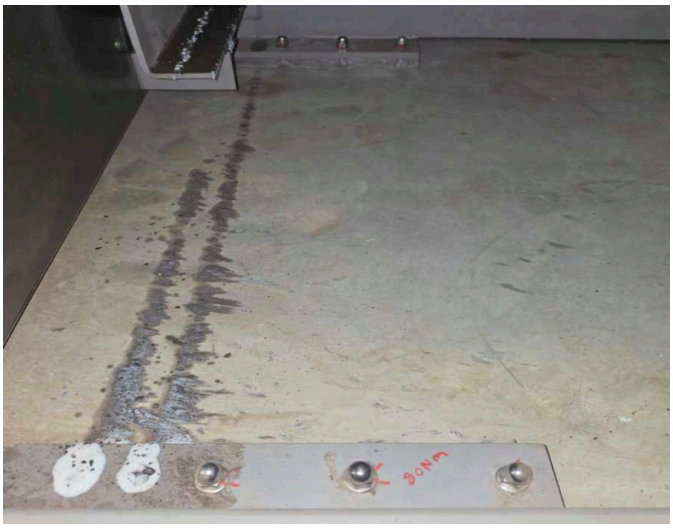



19. Pneumatic Inspection





## 21. Idle Terminal Anchors



### 3. Oven Unloader Maintenance

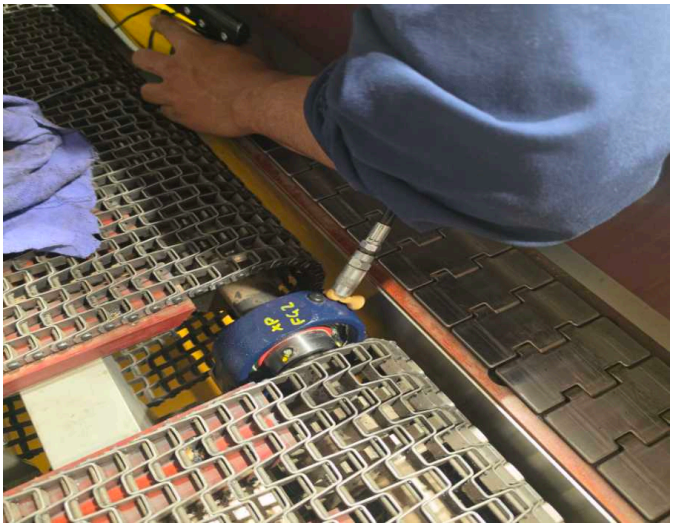
| SN | Inspection Item | Note | Good | Repair | Replace | NA |
|---|---|---|---|---|---|---|
| 22 | Unloader Gearmotors | - 6-12 monthly gearbox oil change (Accelerator & Cross Conv).<br>- Inspect mountings & torque arm bushes.<br>- Noise & vibration Inspection. | ✓ | ○ | ○ | ○ |
| 23 | Unloader Accelerator Conv Bearings | - Accelerator conveyor bearings re-grease (PF XP FG2).<br>- Belt & rod inspection.<br>- Sprocket tooth & alignment inspection. | ✓ | ○ | ○ | ○ |
| 24 | Accelerator Conveyor Drive System | - Chain tension, sprocket tooth inspection for wear, alignment.<br>- Inspect condition of roll ring. | ✓ | ○ | ○ | ○ |
| 25 | Accelerator Motor Cooling Fan | - Verify operation of cooling fan. | ✓ | ○ | ○ | ○ |

| | | | | | | |
|---|---|---|---|---|---|---|
| 26 | Cross Conveyor | - Slat belt & wearstrip inspection.<br>- Drive & idle sprocket inspection.<br>- Roller inspection.<br>- Bearing inspection & re-grease. | ✓ | ○ | ○ | ○ |
| 27 | Cross Conveyor Pneumatic Cylinders | - Mounting, flow control, leak detection, reed switch inspection.<br>- Inspect for smooth, level actuation. | ✓ | ○ | ○ | ○ |
| 28 | Pneumatic Inspection | - Inspect for leaks<br>- Clean ASU filter, bowl, condensate trap. | ✓ | ○ | ○ | ○ |
| 29 | Electrical Inspection | - Inspect loader batch proxies, terminal position proxies, pan position proxies, cross sensor proxy, outfeed panel control operation. | ✓ | ○ | ○ | ○ |
| 30 | Dischrage Terminal Anchors | Inspect terminal anchors for tightness. | ✓ | ○ | ○ | ○ |

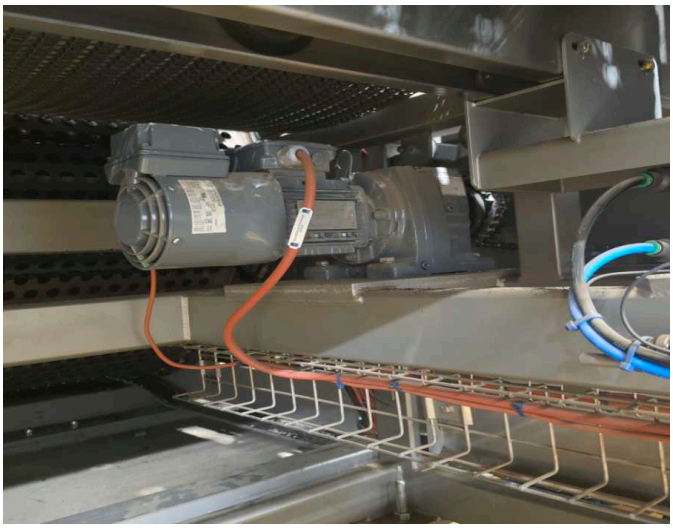
## 22. Unloader Gearmotors



## 23. Unloader Accelerator Conv Bearings

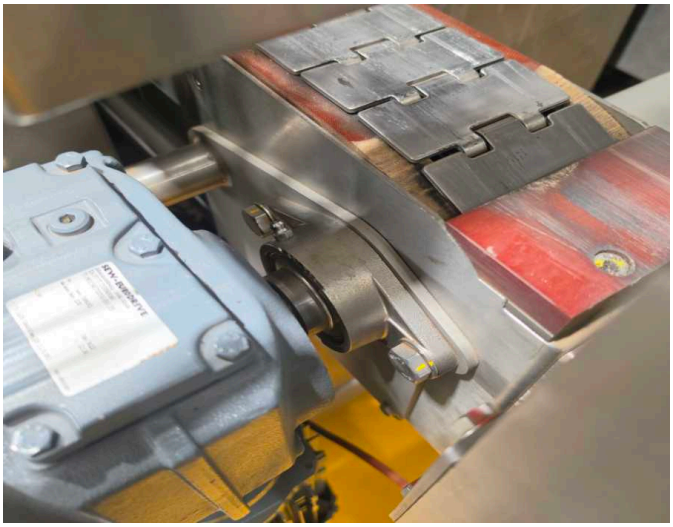


## 25. Accelerator Motor Cooling Fan



26. Cross Conveyor



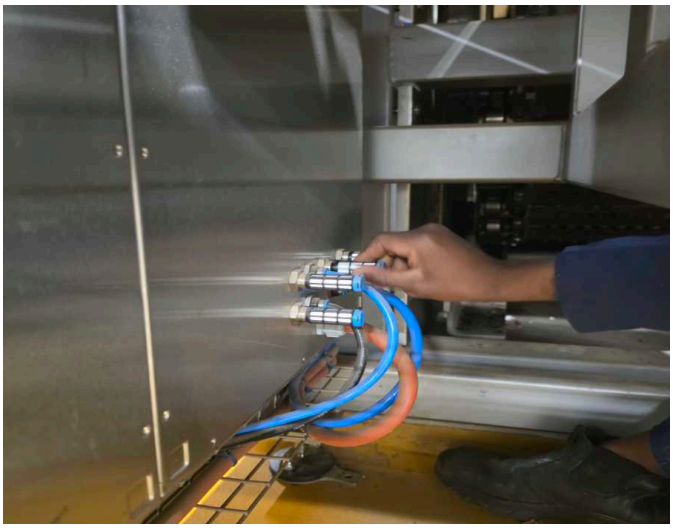


27. Cross Conveyor Pneumatic Cylinders



28. Pneumatic Inspection

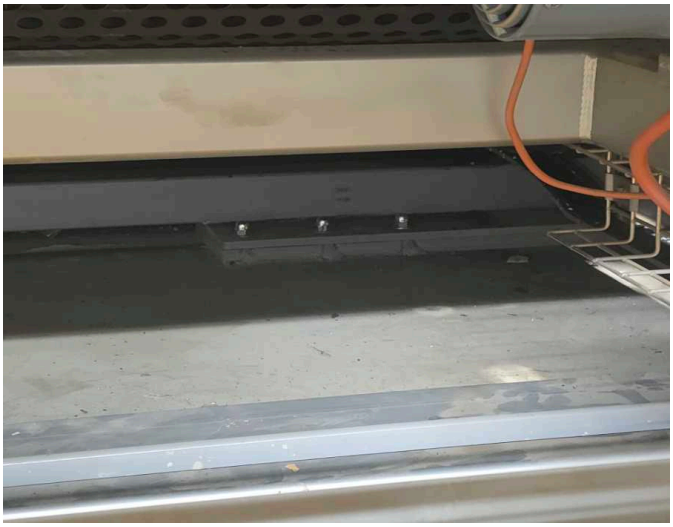




29. Electrical Inspection



30. Dischrage Terminal Anchors

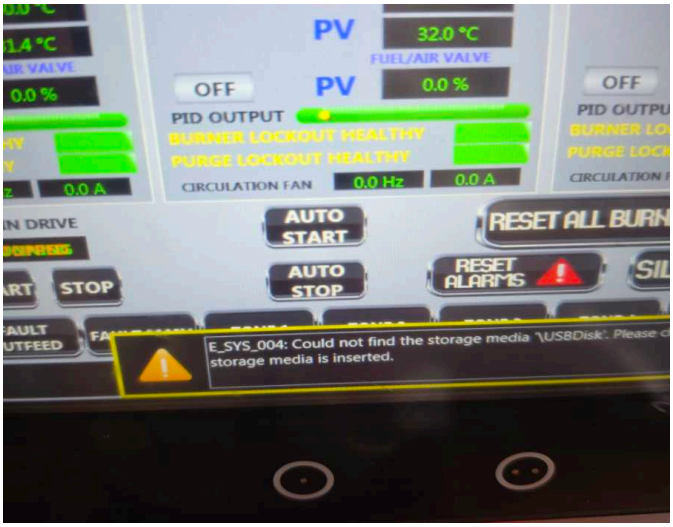


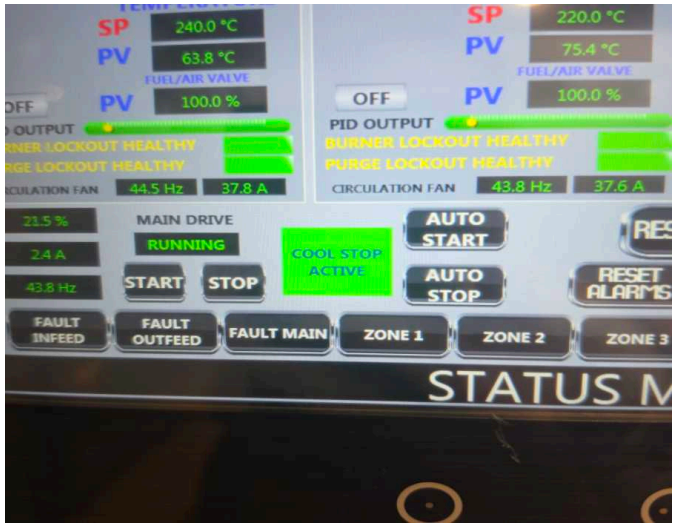
4. Operation

| | | | |
|---|---|---|---|

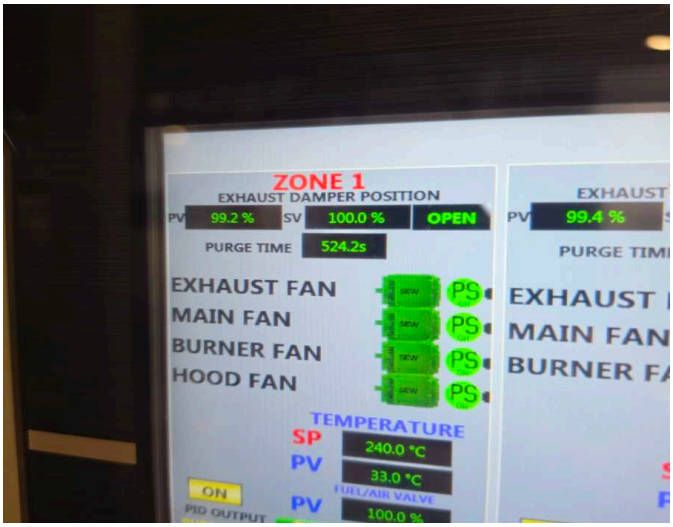
| SN | Inspection Item | Note | Good | Repair | Replace | NA |
|---|---|---|---|---|---|---|
| 31 | Auto Start Cycle | - Pre-startup inspections<br>- Auto Start sequence.<br>- Slow Heat procedure.<br>- Autostart of oven conveyor main drive. | ✓ | ○ | ○ | ○ |
| 32 | Startup & Purge Cycle | - Verification of purge circuit<br>- Auto start of purge cycle and purge countdown.<br>- Troubleshooting unhealthy purge circuit.<br>- Slow Heat & Cool Stop Cycle | ✓ | ○ | ○ | ○ |
| 33 | Basic HMI operation | - Auto Start/Auto Stop<br>- Temperature control.<br>- Exhaust damper control.<br>- Bake Time control.<br>- Overview screens.<br>- Maintenance screens.<br>- Main Drive, Loader & Unloader settings pages<br>- Oven motor block page.<br>- Alarm & alarm history.<br>- Reset, Reset Burners/purge function. | ✓ | ○ | ○ | ○ |
| 34 | Alarms & Troubleshooting | - Common alarm descriptions, troubleshooting & reset procedures (Ongoing process of training during production).<br>- Alarm & alarm history page. | ✓ | ○ | ○ | ○ |
| 35 | Burners | - Basic operating principal of burners.<br>- Gas supply shut-off location, E-Stop, combustion blower isolation, burner local control.<br>- LMV5 Safety PLC.<br>- Pilot gas/diesel switchover (Training to be recieved during oil commissioning). | ✓ | ○ | ○ | ○ |
| 36 | Fortress Lockout System | - Functionality of Loader Fortress lockout.<br>- Key switch panel & loader access procedure.<br>- Start/Stop/Reset/Silence button function.<br>- Load Part Batch function.<br>- Manual home function. | ✓ | ○ | ○ | ○ |
| 37 | Delivery End Manual Control | - Start/Stop main drive function.<br>- Reset/Silence alarm function.<br>- Manual accelerate function.<br>- Manual offload cross conveyor fucntion.<br>- E-Stop. | ✓ | ○ | ○ | ○ |
| 38 | Panel | - Basic description of oven electrical panel & components.<br>- Physical purge circuit<br>- Panel aircon/vent fan inspection. | ✓ | ○ | ○ | ○ |
| 39 | Unloader Operational Sequence | - Pan detection photocells and acceleration sequence, cross conveyor batch detection & unload sequence. | ✓ | ○ | ○ | ○ |
| 40 | Loader Operational Sequence | - Pan batching and load sequence.<br>- Part batch loading. | ✓ | ○ | ○ | ○ |

## 31. Auto Start Cycle

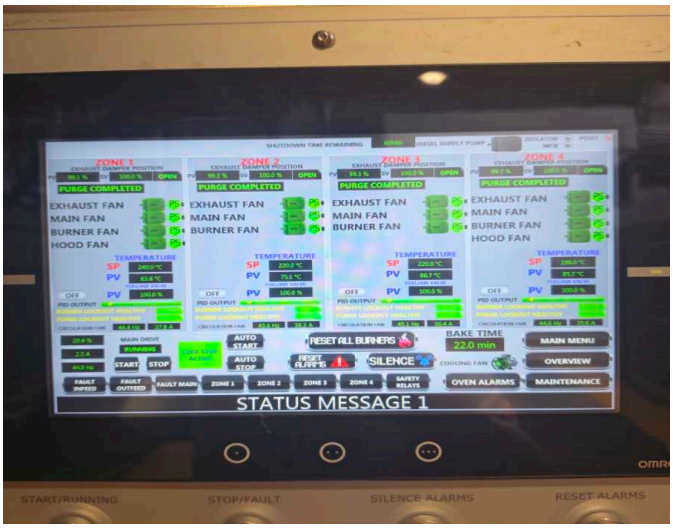


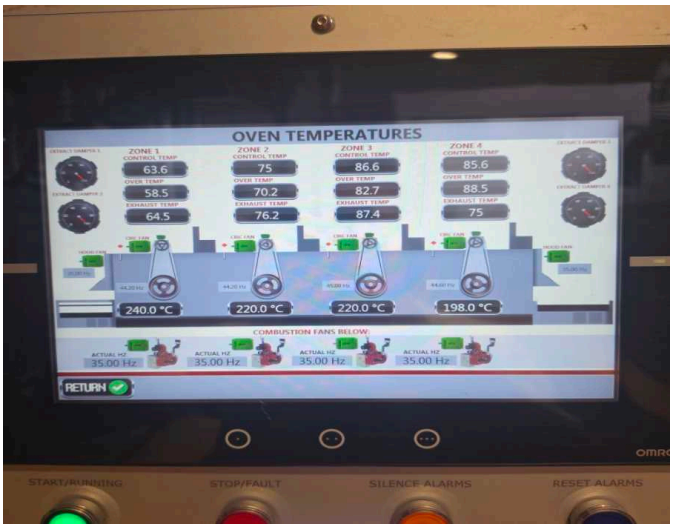

## 32. Startup & Purge Cycle




## 33. Basic HMI operation



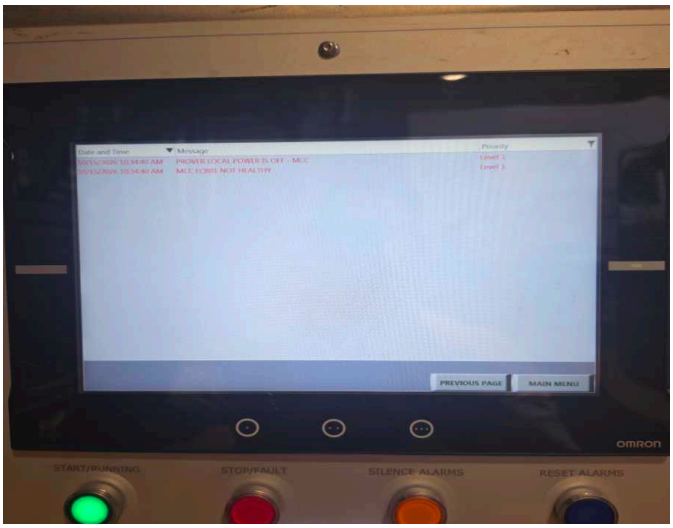


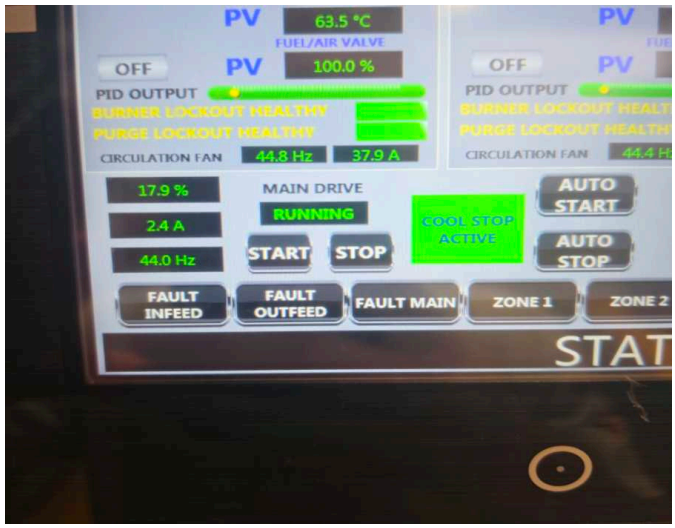





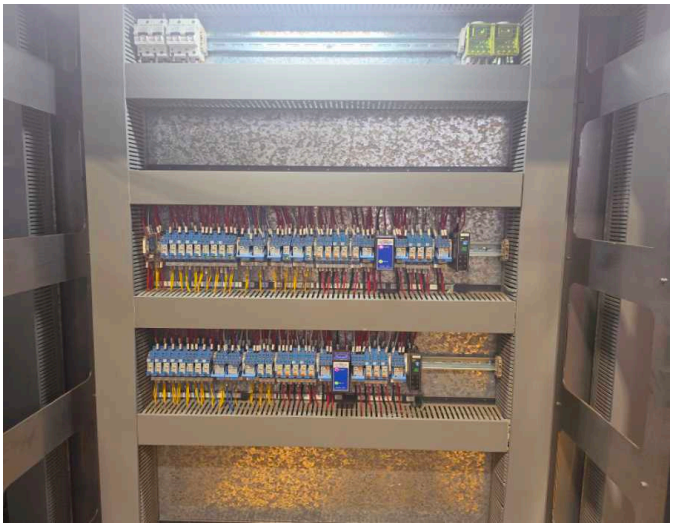


## 34. Alarms & Troubleshooting

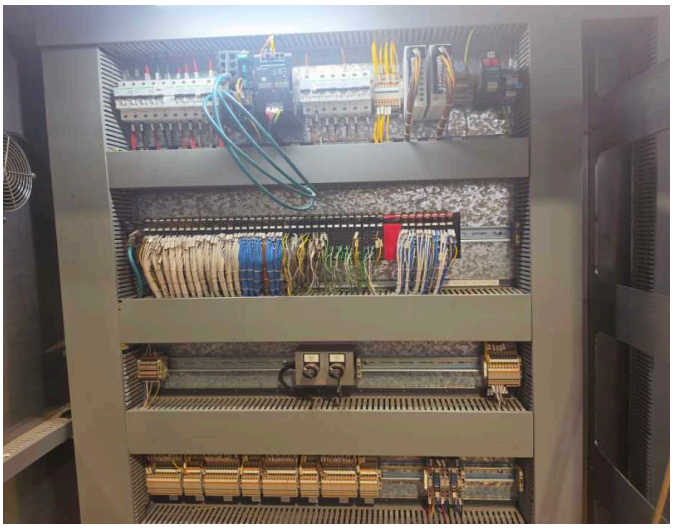



## 38. Panel





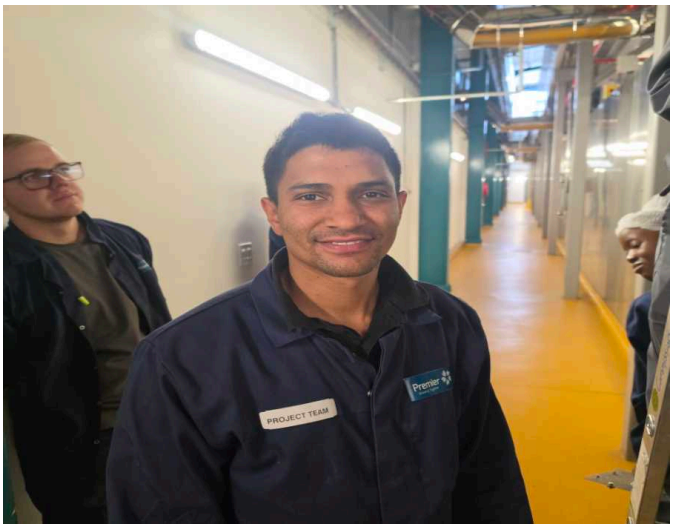


## 5. Training Register

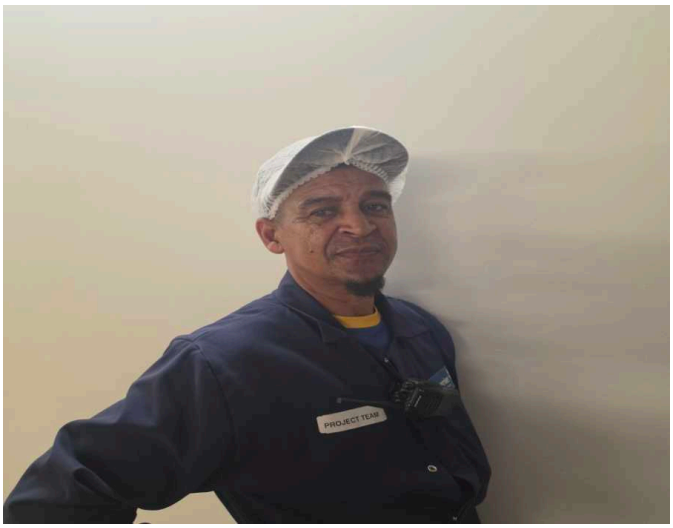
| SN | Inspection Item | Note | Good | Repair | Replace | NA |
|---|---|---|---|---|---|---|
| 41 | Signed Training Register | | ✓ | ○ | ○ | ○ |

| SN | Inspection Item | Note | Pass | Fail | NA |
|---|---|---|---|---|---|
| 42 | Trainee 1 | Wasfiq Theunissen | ✓ | ○ | ○ |
| 43 | Trainee 2 | Delmaine Fred | ✓ | ○ | ○ |
| 44 | Trainee 3 | Tumisho Molefe | ✓ | ○ | ○ |
| 45 | Trainee 4 | Piet Matjomane | ✓ | ○ | ○ |

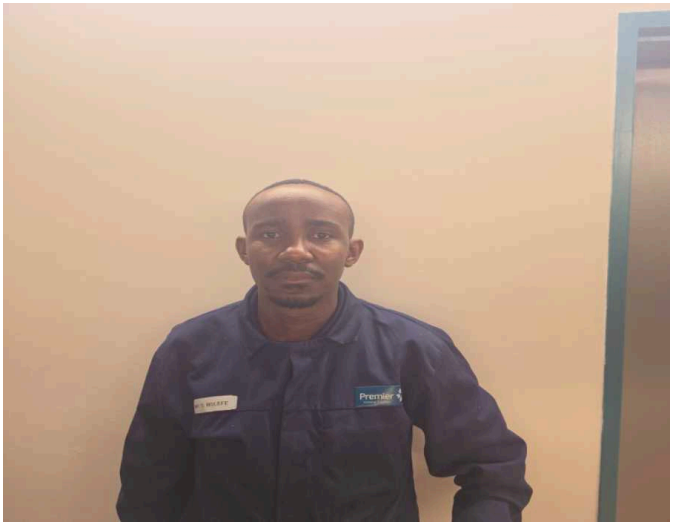
### 42. Trainee 1




### 43. Trainee 2




44. Trainee 3




45. Trainee 4



| Additional Note |
| --- |
| Disclaimer: This document serves as an overview of training conducted with the Blue Ribbon Middelburg Maintenance team. This is not the official maintenance or operation manual of the machine, and serves as a quick guide for staff to reference in future. |

| | |
| --- | --- |
| **Overall Condition:** | Excellent Condition |
| **Asset Safe to Use:** | **Yes** |
| **Maintenance Required:** | No |
| **Asset Status** | Available |

## Custom Business Fields

Client Name : BR Maintenance Team

Contact details : 403-5



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**Dale Technician Signature**



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**Client Signature**